

Ameer Dharamshi

ASSISTANT PROFESSOR · MATHEMATICS AND STATISTICS

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Education

University of Washington

PHD IN BIOSTATISTICS

• Advisors: Daniela Witten and Jon Wakefield

Seattle, WA, USA

September 2022 - June 2026

University of Toronto

MSC IN STATISTICAL SCIENCES

Toronto, ON, Canada

September 2019 - June 2020

University of Waterloo

BMATH IN MATHEMATICS/CPA

• Minors in Statistics and Computer Science

Waterloo, ON, Canada

September 2013 - June 2018

Professional Experience

2026-Present **Assistant Professor (tenure-track)**, Department of Mathematics and Statistics, McGill University
2024-Present **Research Consultant**, Department of Demography, University of California Berkeley
2022-2026 **Graduate Research Assistant**, Department of Biostatistics, University of Washington
2020-2022 **Research & Data Consultant**, Global Education Monitoring Report, UNESCO
2021-2022 **Part-time Researcher**, Bayesian Demography Lab, University of Toronto
2018-2019 **Consultant**, Deloitte Canada

Publications

PUBLISHED

Wu, Y., **Dharamshi, A.**, & Wakefield, J. (2026). Small Area Estimation of Education Levels in Low- and Middle-Income Countries. *Annals of Applied Statistics*, 20(1), 833-855. Available from: <https://doi.org/10.1214/25-AOAS2135>.
Dharamshi, A., Neufeld, A., Gao, L. L., Witten, D., & Bien, J. (2026). Thinning a Wishart Random Matrix. *Biometrika*, 113(1). Available from: <https://doi.org/10.1093/biomet/asaf081>.
Dharamshi, A., Neufeld, A., Gao, L. L., Bien, J., & Witten, D. (2026). Decomposing Gaussians with Unknown Covariance. *Biometrika*, 113(1). Available from: <https://doi.org/10.1093/biomet/asaf057>.
Neufeld, A., **Dharamshi, A.**, Gao, L. L., Witten, D., & Bien, J. (2025). Discussion of “Data fission: splitting a single data point”. *Journal of the American Statistical Association: Theory & Methods*, 120(549), 151-157. Available from: <https://doi.org/10.1080/01621459.2024.2421998>.
Dharamshi, A., Neufeld, A., Motwani, K., Gao, L. L., Witten, D., & Bien, J. (2025). Generalized data thinning using sufficient statistics. *Journal of the American Statistical Association: Theory & Methods*, 120(549), 511-523. Available from: <https://doi.org/10.1080/01621459.2024.2353948>.
Dharamshi, A., Alexander, M., Winant, C., & Barbieri, M. (2025). Jointly estimating subnational mortality for multiple populations. *Demographic Research*, 52, 71-110. Available from: <https://doi.org/10.4054/DemRes.2025.52.3>.
Neufeld, A., **Dharamshi, A.**, Gao, L. L., & Witten, D. (2024). Data thinning for convolution-closed distributions. *Journal of Machine Learning Research*, 25(57), 1-35. Available from: <https://doi.org/10.48550/arXiv.2301.07276>.
Dharamshi, A., Ngo, V., & Rosenthal, J. S. (2024). Sampling by divergence minimization. *Communications in Statistics - Simulation and Computation*, 53(12), 6071-6095. Available from: <https://doi.org/10.1080/03610918.2023.2199352>.

Dharamshi, A., Barakat, B., Alkema, L., & Antoninis, M. (2022) A Bayesian model for estimating Sustainable Development Goal indicator 4.1.2: School completion rates. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 71(5), 1822–1864. Available from: <https://doi.org/10.1111/rssc.12595>.

PREPRINTS

Dharamshi, A., Zou, R., & Witten, D., (2026). Testing hypotheses via orthogonalization. *ArXiv preprint*. Available from: <https://arxiv.org/abs/2606.29732>.

Dharamshi, A., Richards, M., Childress, S., Lee, B., & Casey, D., (2026). Estimating Residential Displacement in the Central Puget Sound Region using Household Survey Data. *ArXiv preprint*. Available from: <https://arxiv.org/abs/2603.06437>.

Dharamshi, A., Gao, P., & Wakefield, J. (2025). Exact variance estimation for model-assisted survey estimators using U- and V-statistics. *ArXiv preprint*. Available from: <https://arxiv.org/abs/2502.11032>.

Dharamshi, A., Antoninis, M., Montoya, S., & Barakat, B. (2023). A Bayesian cohort model for estimating out-of-school rates and populations. *SocArXiv preprint*. Available from: <https://osf.io/preprints/socarxiv/sqwb2/>

Scholarships & Awards

2026-2029	Start-up Funding , McGill University	150,000 CAD
2026	Thomas R. Fleming Excellence in Biostatistics Award , University of Washington	1,000 USD
2026-2028	NSERC CPRA (Declined) , Natural Sciences and Engineering Research Council of Canada	140,000 CAD
2024-2027	NSF-HNDS - Senior Investigator (PI: Magali Barbieri, University of California Berkeley) , National Sciences Foundation	791,403 USD (49,500 to AD)
2024	Student Paper Award , JSM Statistical Learning and Data Science Section	1,000 USD
	CSSS Travel Award (EPC 2024) , University of Washington	500 USD
2023-2026	NSERC PGS D , Natural Sciences and Engineering Research Council of Canada	96,250 CAD
2022	Top Scholar Recruitment Award , University of Washington	7,500 USD
	Faculty of Arts and Science Doctoral Recruitment Award (Declined) , University of Toronto	5,000 CAD
2013-2017	Faculty of Mathematics Entrance Scholarship , University of Waterloo	5,000 CAD
2013	President's Scholarship of Distinction , University of Waterloo	5,000 CAD

Presentations

INVITED TALKS

Testing hypotheses via orthogonalization

1. Bocconi University (February 2025)

Expanding the scope of post-selection inference

1. McGill University (December 2025)
2. ETH Zürich (December 2025)
3. University of Toronto (December 2025)
4. NUS-IMS Young Mathematical Scientists Forum - Statistics and Data Science 2025 (November 2025)
5. University of Cambridge (November 2025)

Estimating Residential Displacement in Puget Sound using Household Survey Data

1. Public Health Seattle & King County (June 2025)

Data thinning with applications in the health sciences

1. SSC Annual Meeting 2024 (April 2024)

Generalized data thinning using sufficient statistics

1. JSM 2024 (August 2024)
2. G-Research ML College Talk (August 2023)
3. International Seminar on Selective Inference (April 2023)

Estimating SDG 4 Indicators with a Bayesian Modelling Framework: Completion and Out-of-School Rates

1. 9th Annual Meeting of the Technical Cooperation Group on SDG4 Indicators: Joint EMIS and Surveys Working Group (November 2022)

A Bayesian Cohort Model for Estimating SDG Indicator 4.1.4: Out-of-School Rates

1. STAB Working Group, University of Washington (November 2022)
2. Bayes Working Group, University of Massachusetts Amherst. (December 2021)
3. 8th Annual Meeting of the Technical Cooperation Group on SDG4 Indicators (TCG): Household Surveys Working Group (October 2021)

Adjusted Bayesian Completion Rates (ABC)

1. Bayes Working Group, University of Massachusetts Amherst (June 2021)

CONTRIBUTED PRESENTATIONS

Testing hypotheses via orthogonalization

1. JSM 2026 (August 2026)

Estimating subnational fertility rates with a principal component-based Bayesian spatiotemporal model

1. International Population Conference 2025 (July 2025)

Estimating the Composition of the Cystic Fibrosis Pulmonary Microbiome

1. North American Cystic Fibrosis Conference 2024 (September 2024)

A Principled Approach to Model Validation in the Context of Estimating Age-Specific Rates

1. European Population Conference 2024 (Poster) (June 2024)

A Bayesian Hierarchical Model for Jointly Estimating Subnational Mortality for Multiple Populations

1. PAA Annual Meeting. (April 2022)
2. 6th HMD Symposium (June 2022)
3. European Population Conference 2022 (June 2022)

CSEye: A Proposed Solution for Accurate and Accessible One-to-Many Face Verification

1. AAAI-2019 (Poster) (January 2019)

Teaching Experience _____

McGill University

Fall 2026 **MATH 682: Statistical Inference**, Instructor

University of Washington

Spring 2024 **DATA 558: Statistical Machine Learning for Data Scientists**, Teaching Assistant

Winter 2024 **BIOST 515: Biostatistics II**, Teaching Assistant

University of Toronto

Summer 2020 **STA 302: Methods of Data Analysis I**, Teaching Assistant
Winter 2020 **STA 238: Probability, Statistics and Data Analysis II**, Teaching Assistant
Fall 2019 **STA 237: Probability, Statistics and Data Analysis I**, Teaching Assistant

University of Waterloo

Summer 2015 **MATH 127: Calculus 1 for the Sciences**, Teaching Assistant
Fall 2014 **MATH 135: Algebra for Honours Mathematics**, Teaching Assistant

Outreach & Professional Development _____

PEER REVIEW

Journal of the American Statistical Association - Theory & Methods, Electronic Journal of Statistics, Demographic Research, Population Health Metrics, Journal of Survey Statistics and Methodology, Survey Methodology, Biostatistics

COMMUNITY SERVICE

2026-Present **Statistics Seminar Organizer**, McGill University
2026-Present **Committee of Graduate Affairs Member**, McGill University
2025-2026 **Space Time Analysis Bayes (STAB) Research Group Organizer**, University of Washington
2024-2026 **Biostatistics Student Seminar Organizer**, University of Washington
2023-2026 **EPTEC Committee Member (Biostatistics Curriculum)**, University of Washington
2023-2024 **PARS Reviewer**, University of Washington
2022-2023 **Biostatistics EDI Committee Member**, University of Washington

Additional Information _____

- Technical Skills: R/RMarkdown, Stan, INLA, Python, SQL, C/C++, MATLAB, Julia
- General Skills: LaTeX and Microsoft Office
- Languages: English (native), French (basic)
- Hobbies: reading, squash, tennis, skating, baking